

cfRNA-BrainTrace Supplementary Figures and Tables

Updated for the 2026-06-19 three-tier hybrid validation route

Includes copied validation figures and regenerated supplementary tables.

Figure1 three tier validation summary

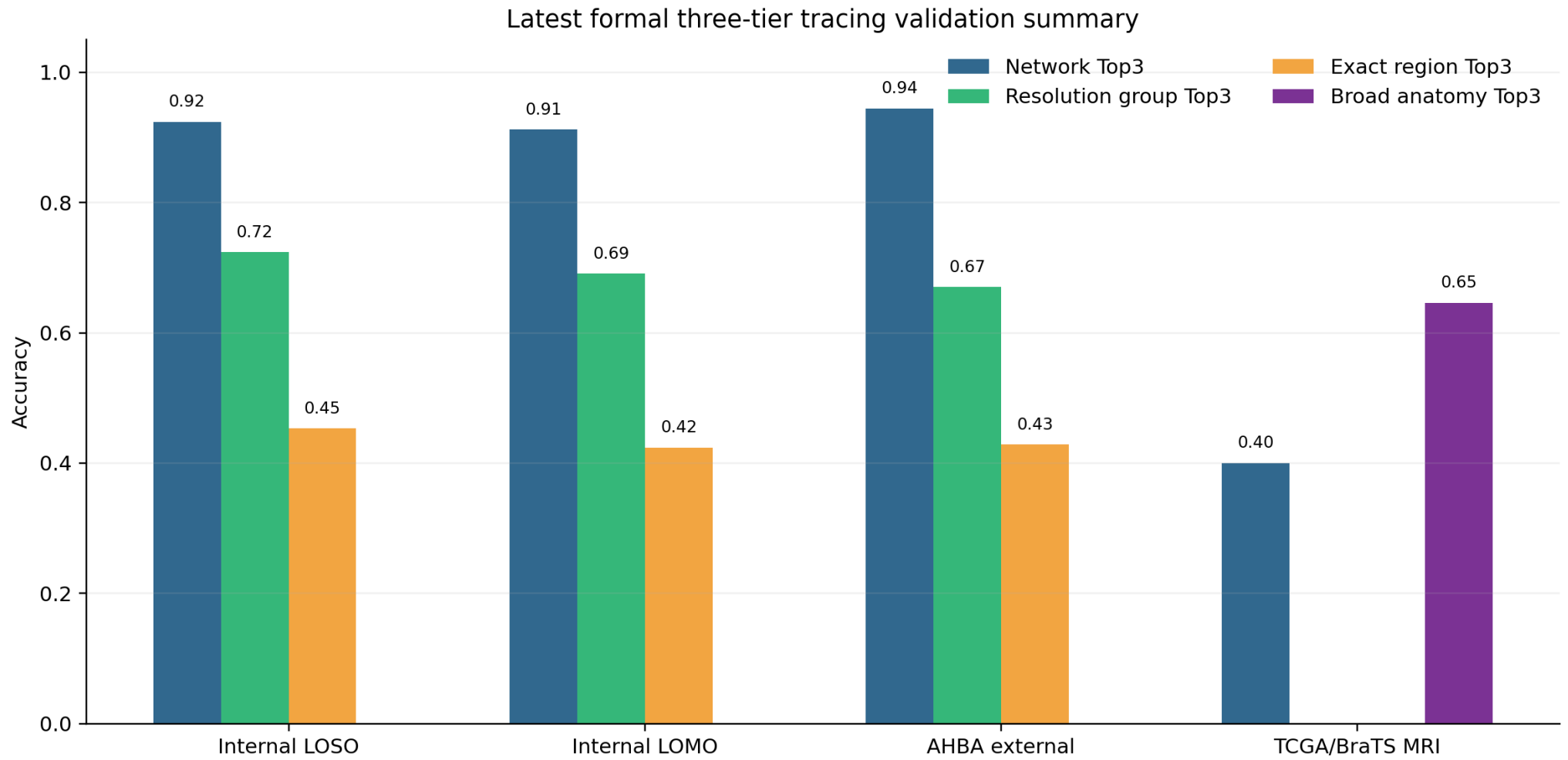


Figure 1. Three-tier validation summary for projected-VSD Network beam and logCPM downstream reranking.

FigureS1 projector gene qc distributions

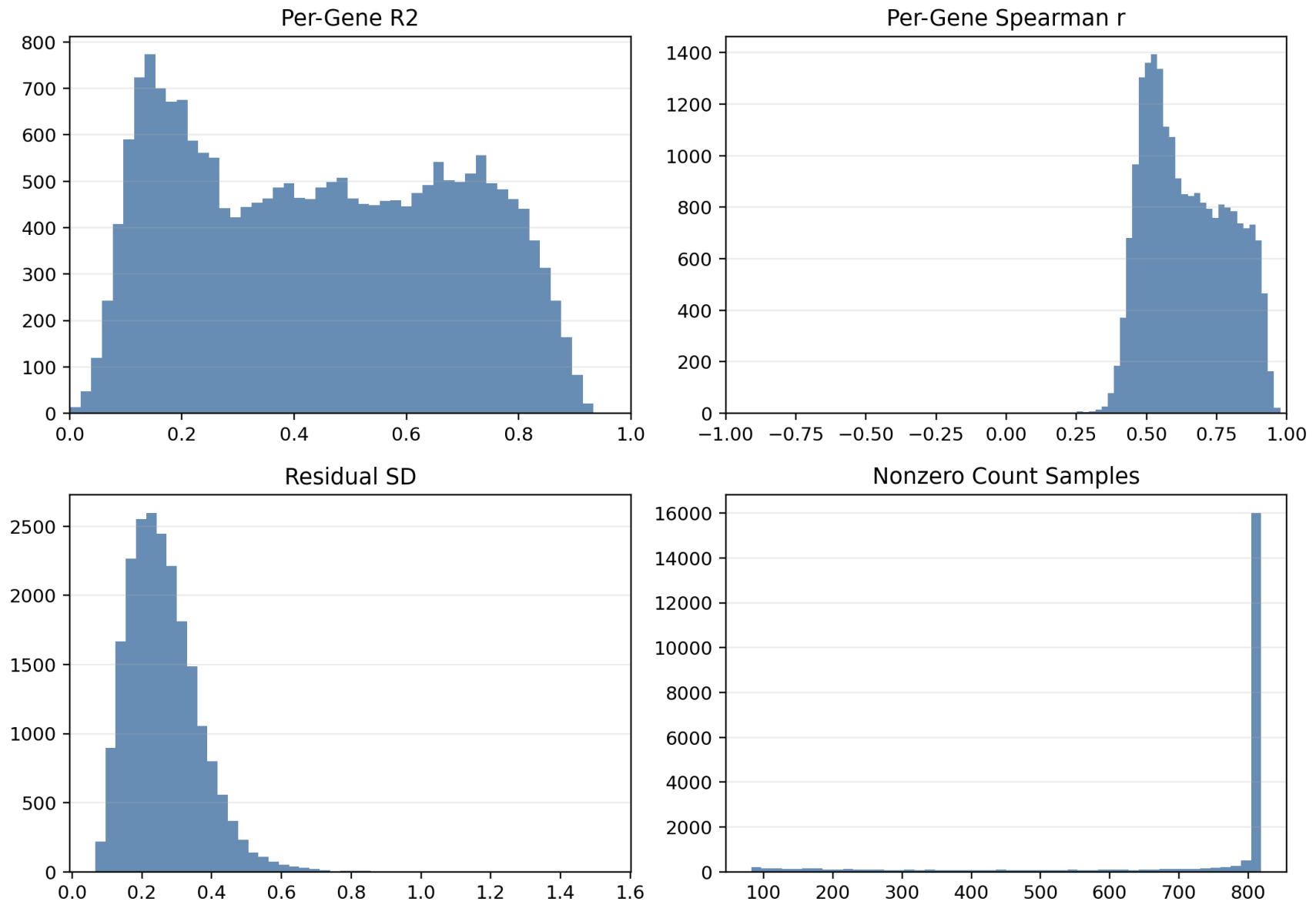


Figure S1. Projector gene-level and sample-level QC distributions.

FigureS2 internal network accuracy

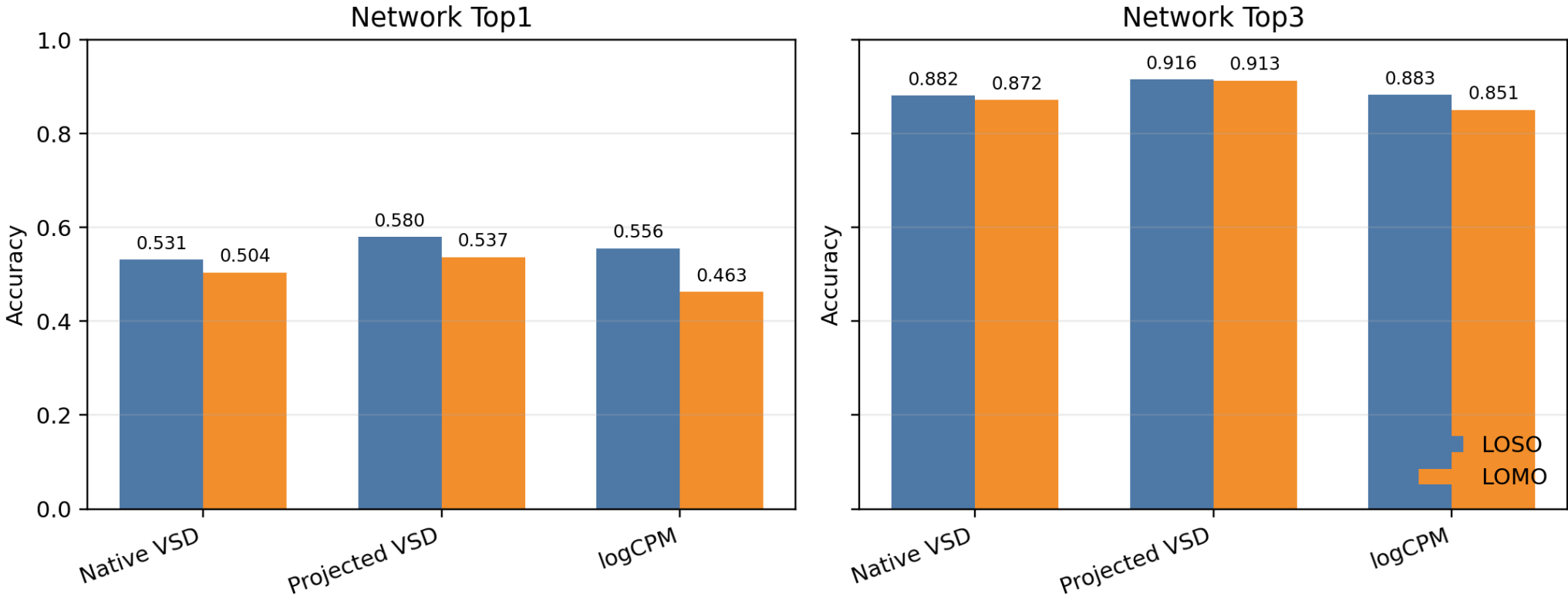


Figure S2. Internal Network accuracy comparing logCPM, native VSD and projected VSD routes.

FigureS3 AHBA formal three tier accuracy

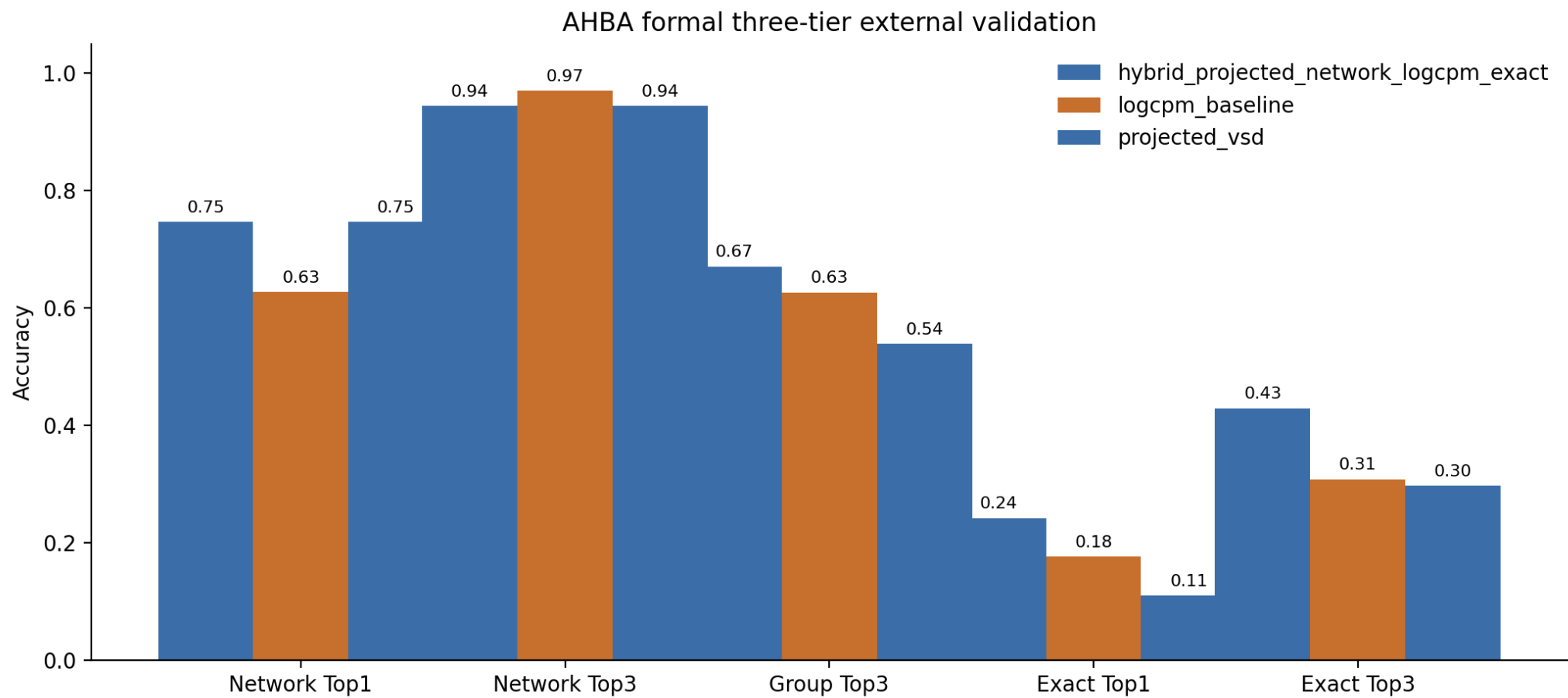


Figure S3. AHBA formal three-tier mapped-label external validation.

FigureS4 TCGA BraTS coarse consistency

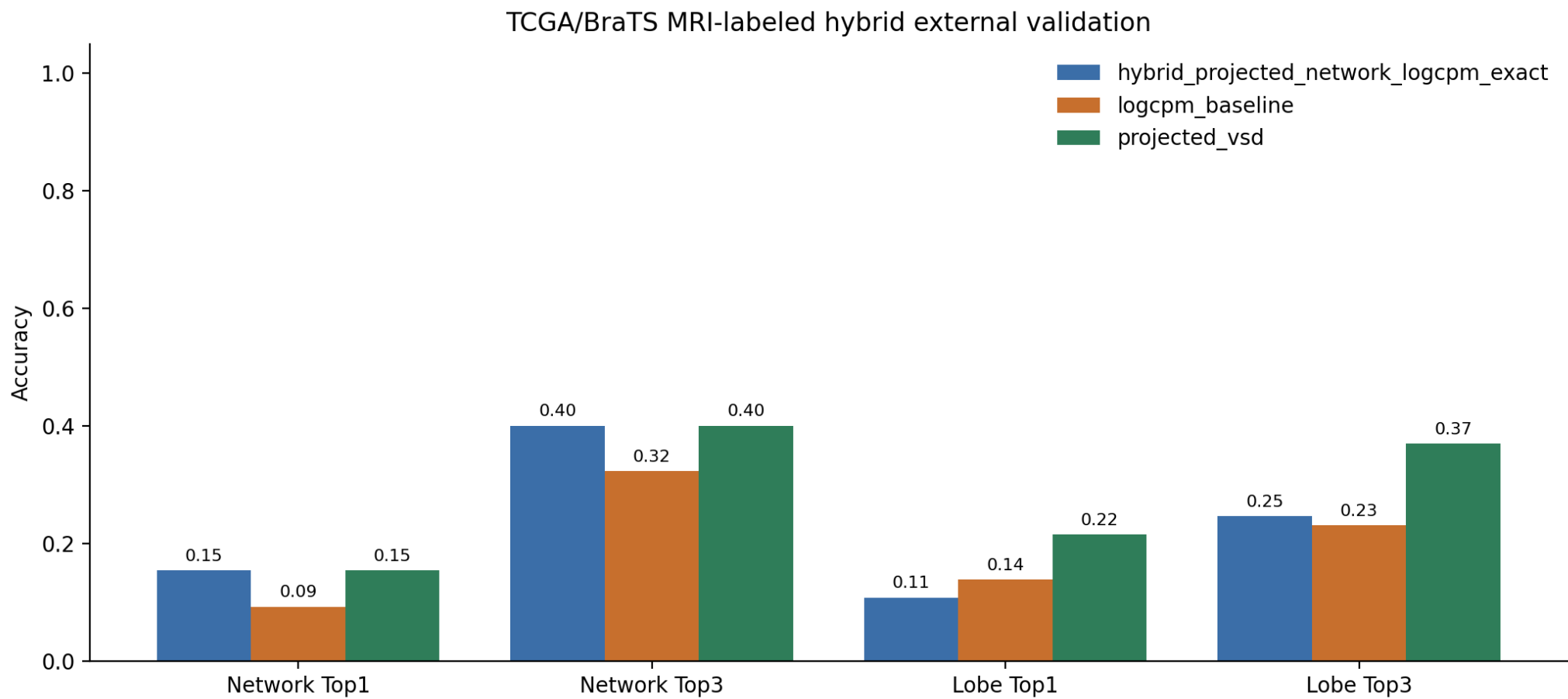


Figure S4. TCGA/BraTS glioma RNA-seq and MRI-label coarse anatomical consistency.

Table1 route components

Component	Implementation	Claim boundary
Projector	Query logCPM/logTPM-compatible expression -> Bo2023-like VSD	Network beam only; not a new atlas
Network	Projected-VSD Top3 candidate generation	Primary endpoint
Resolution group	logCPM local rerank within Network beam	Main region-level endpoint
Exact region	logCPM local candidate rerank	Exploratory local ranking
External validation	AHBA mapped labels; TCGA/BraTS coarse labels	AHBA mapped-label support; TCGA/BraTS coarse consistency only

TableS1 projector gene audit

Metric	Value	Interpretation
Common Bo2023 samples	819	Counts, VSD and metadata aligned
Common genes after symbol mapping	21668	Projector gene space
Locked Network genes in common panel	199/200	One prior date-like symbol issue cleaned
Projector global Pearson	0.9961	Strong sample-level VSD-like reconstruction
Projector median sample Pearson	0.9969	Supports Network beam use
Projector median gene Pearson	0.6619	Do not claim gene-perfect reconstruction
Date-like symbols after cleaning	0	Excel-mangled symbols removed
Ensembl fallback symbols	5499	Interpretability limitation

TableS2 internal formal validation

Validation	Level	n	Top1	Top3	Median true rank
LOSO formal hybrid	Network	814	0.5835	0.9238	1.0
LOSO formal hybrid	Resolution group	814	0.4447	0.7236	2.0
LOSO formal hybrid	Exact region	814	0.2248	0.4533	4.0
LOMO formal hybrid	Network	819	0.5775	0.9121	1.0
LOMO formal hybrid	Resolution group	812	0.4138	0.6909	2.0
LOMO formal hybrid	Exact region	812	0.2217	0.4236	5.0

TableS3 diagnostic routes

Route	Validation	Level	Top1	Top3	Interpretation
Projected VSD direct	LOS0	Exact region	0.1671	0.3563	Better than logCPM baseline but not enough as main endpoint
Projected VSD direct	LOM0	Exact region	0.1453	0.3116	Below native VSD Top3; diagnostic only
Projected Network + logCPM local	LOS0	Exact region	0.2088	0.4263	Supports downstream local rerank
Projected Network + logCPM local	Formal LOS0	Exact region	0.2248	0.4533	Best formal exact route but still exploratory

TableS4 external validation

Dataset	Route	Endpoint	Top1	Top3	Allowed claim
AHBA	Hybrid	Network	0.7468	0.9442	Cross-species mapped-label support
AHBA	Hybrid	Resolution group	0.3626	0.6703	Group-level mapped-label support
AHBA	Hybrid	Exact region	0.2418	0.4286	Only evaluable mapped labels
TCGA/BraTS	Hybrid	Network	0.1538	0.4000	Coarse anatomical consistency
TCGA/BraTS	Hybrid	Broad anatomy	0.1385	0.6462	Coarse anatomical consistency
GSE189919	Projection feasibility	Gene overlap	15622/21668	0.7210	No accuracy claim

TableS5 claim boundaries

Do not write	Use instead
The whole Bo2023 atlas was reprojected into a new atlas.	Only the query is projected to Bo2023-like VSD for Network beam generation.
Projected VSD is universally superior at exact-region resolution.	Projected VSD supports Network beam generation; group/exact rerank uses logCPM local space.
TCGA/BraTS validates Bo2023 exact-region accuracy.	TCGA/BraTS supports coarse anatomical consistency only.
Caudate exact Top1 performed well.	Caudate/Putamen/Insula are stable at Network/group Top3; Caudate exact Top1 was 0.
Exact Top1 is a deterministic localization result.	Exact outputs are local candidate rankings with clear limits.